

5 C

By considering conservation of energy,
gain in KE = work done by force – work done against frictional forces

$$\frac{1}{2}mv^2 = Fd_{total} + fd_{10}$$

$$\frac{1}{2}(100)v^2 = (320)(30) - (30)(10)$$

$$v = 13.6 \text{ m s}^{-1}$$

6 D